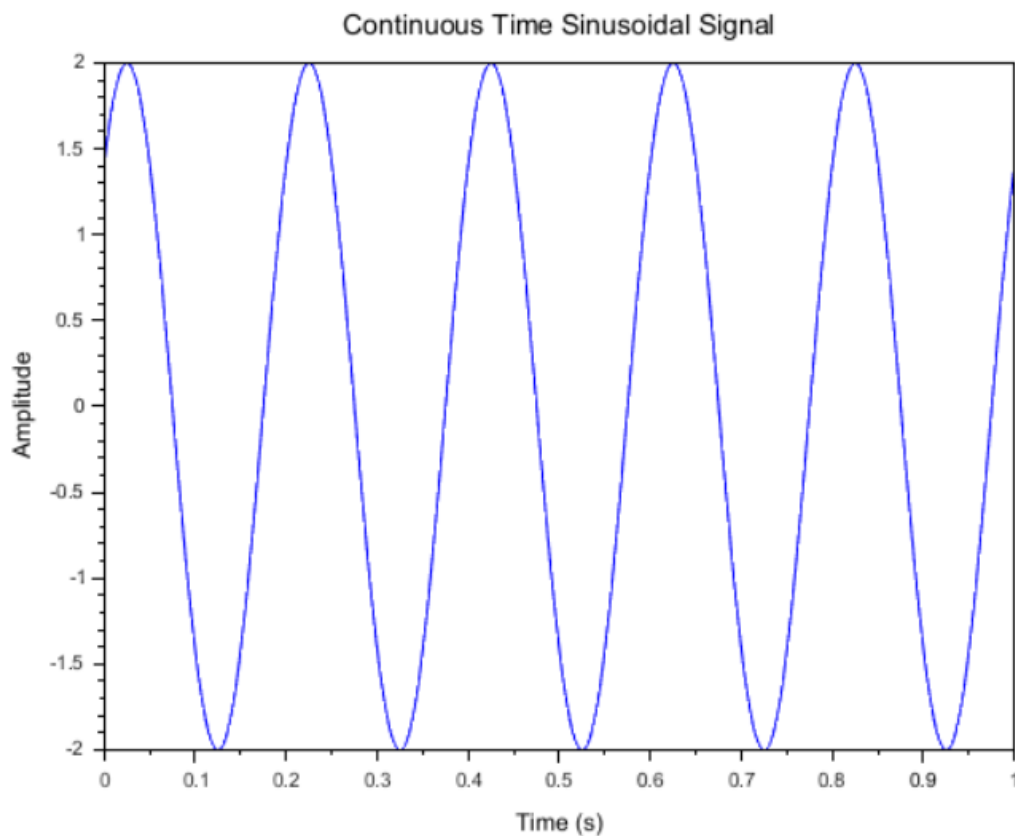


1. SIGNAL GENERATION AND BASIC SIGNAL OPERATIONS

Objective:

```
// Test Case: Sinusoidal Signal Generation
// Define parameters
f = 5; // Frequency in Hz
A = 2; // Amplitude
phi = %pi/4; // Phase shift in radians

t = 0:0.001:1; // Time vector from 0 to 1 second with step size 0.001
// Generate the sinusoidal signal
x_t = A * sin(2 * %pi * f * t + phi);
// Plot the signal
clf; // Clear the current figure
plot(t, x_t);
xlabel('Time (s)');
ylabel('Amplitude');
title('Sinusoidal Signal: 5 Hz, Amplitude 2, Phase Shift  $\pi/4$ ');
xtitle('Continuous Time Sinusoidal Signal','Time (s)','Amplitude');
```



Experiment 1.2

// Test Case: Discrete-Time Unit Impulse Signal Generation

// Define parameters

N = 21; // Length of the signal

n0 = 11; // Index at which the impulse occurs

n = 1:N; // Discrete time index vector

// Generate the unit impulse signal

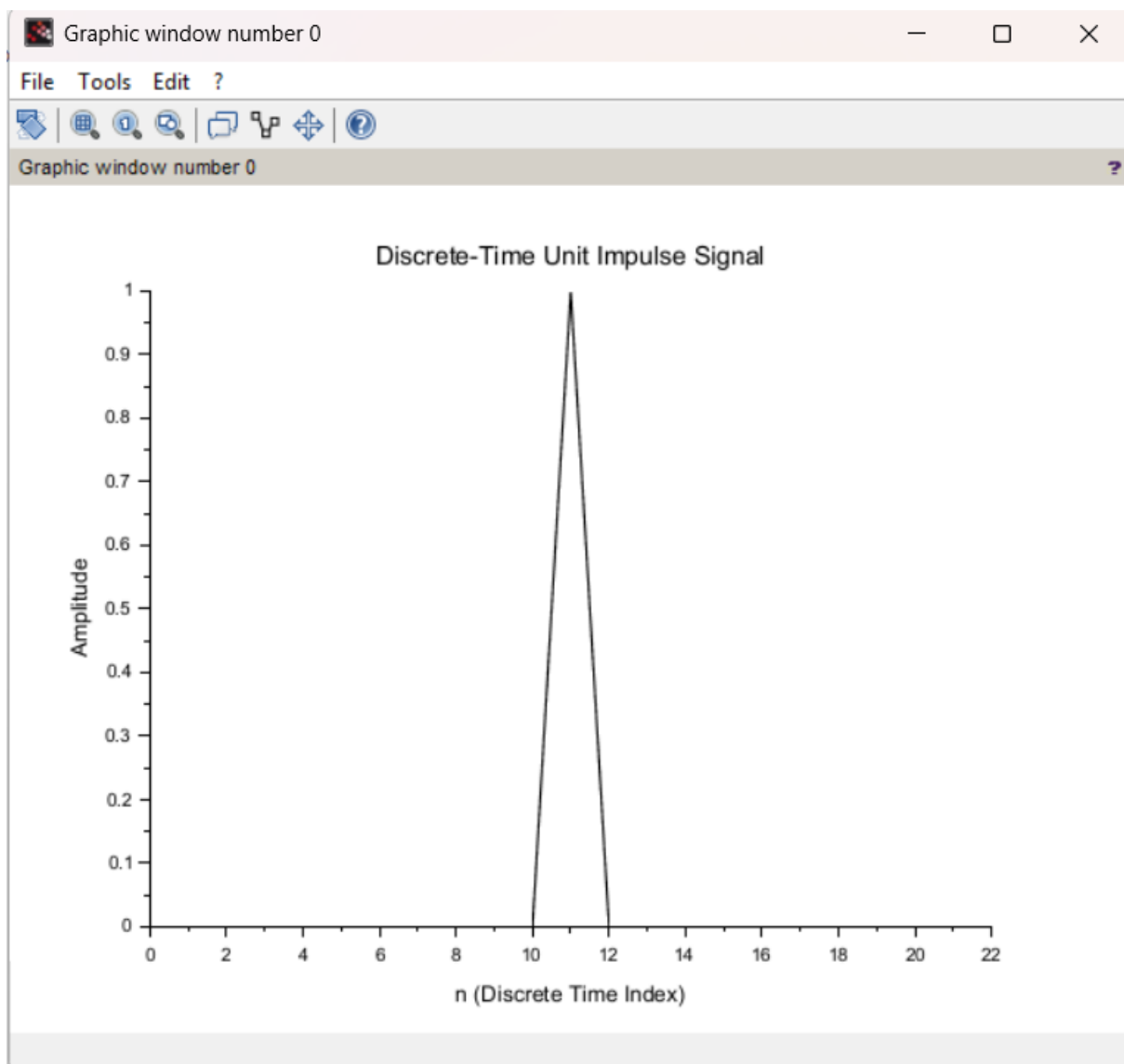
x_n = zeros(1, N); // Initialize signal with zeros

x_n(n0) = 1; // Set the impulse at the desired location

// Plot the signal

clf; // Clear the current figure

```
plot2d(n, x_n);  
xlabel('n (Discrete Time Index)');  
ylabel('Amplitude');  
title('Discrete-Time Unit Impulse Signal');  
xtitle('Discrete-Time Unit Impulse Signal','n (Discrete Time  
Index)','Amplitude');
```



Experiment 1.3

// Continuous time

```
t = -2:0.001:2;
```

```
u = (t >= 0);
```

```
r = t .* (t >= 0);
```

```
imp = (t == 0); // ideal impulse approx is not possible in continuous  
clf;
```

```
subplot(3,1,1);
```

```
plot(t, u);
```

```
xtitle("Unit Step (Continuous)");
```

```
subplot(3,1,2);
```

```
plot(t, r);
```

```
xtitle("Ramp (Continuous)");
```

```
subplot(3,1,3);
```

```
plot(t, imp);
```

```
xtitle("Impulse Approximation");
```

// Discrete time

```
n = -5:5;
```

```
ud = (n >= 0);
```

```
rd = n .* (n >= 0);
```

```
impd = (n == 0);
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot2d3(n, ud); // discrete "stem-like" plot
```

```
xtitle("Unit Step (Discrete)");
```

```
subplot(3,1,2);
```

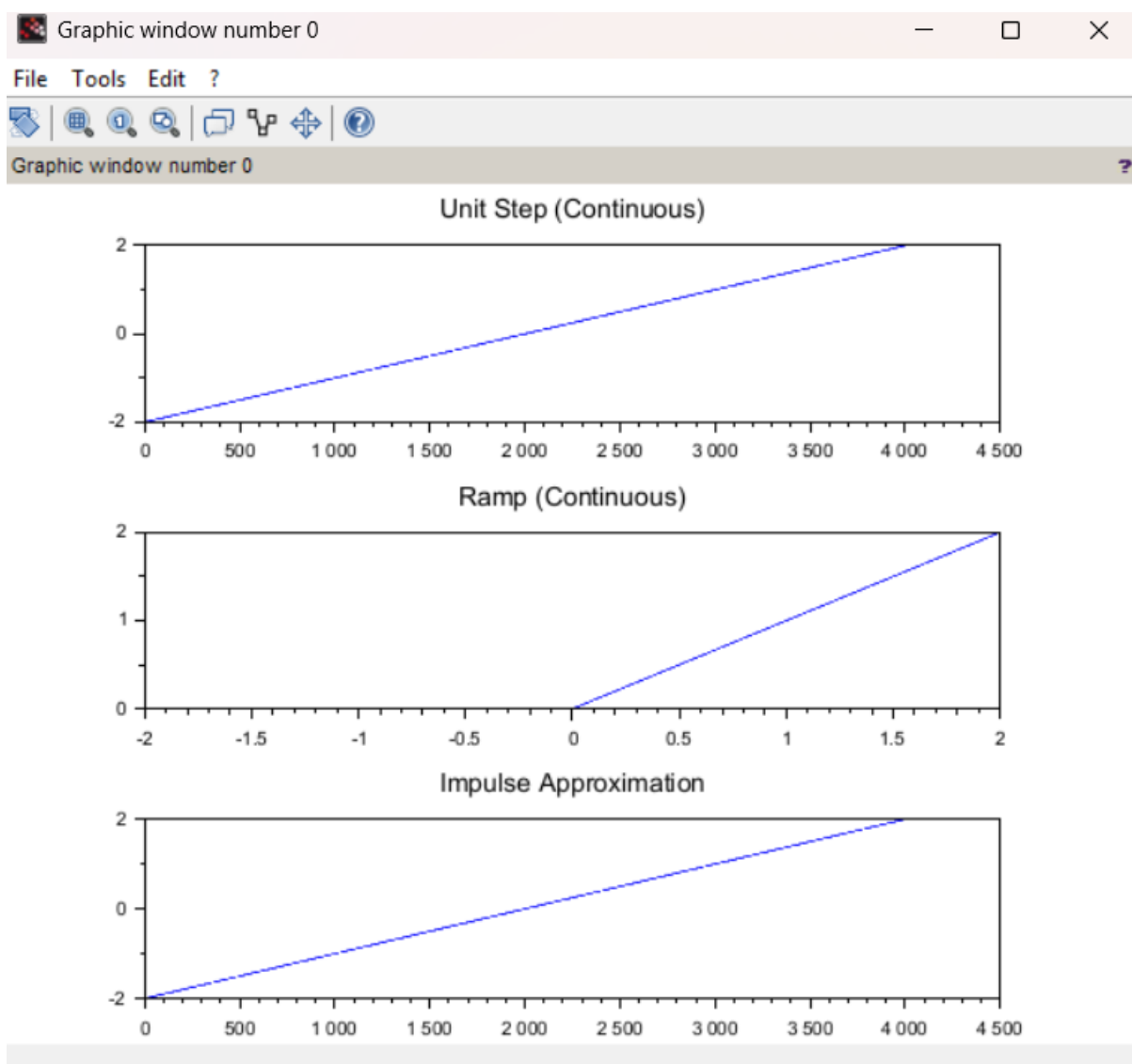
```
plot2d3(n, rd);
```

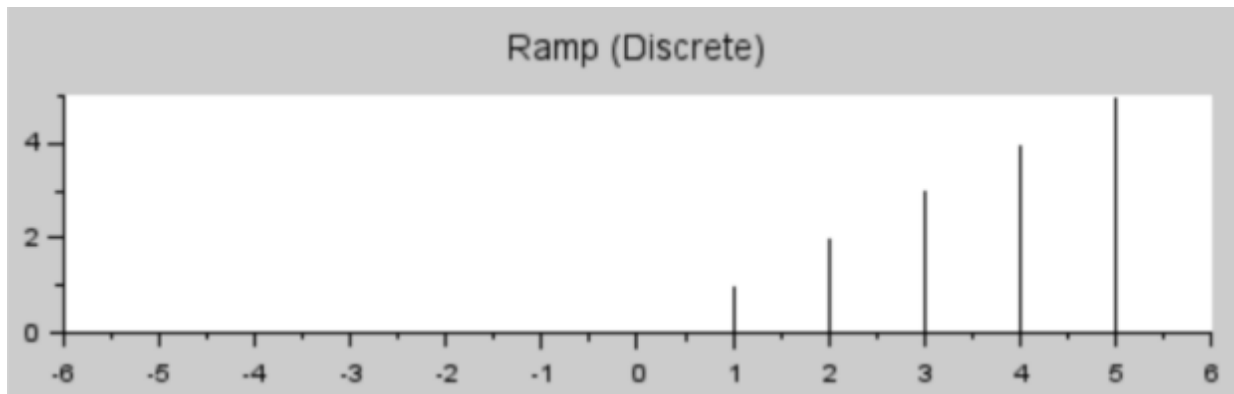
```
xtitle("Ramp (Discrete)");
```

```
subplot(3,1,3);
```

```
plot2d3(n, impd);
```

```
xtitle("Unit Impulse (Discrete)");
```





Experiment 1.4

```
clear; clc;
```

```
// Continuous time
```

```
t = -2:0.001:2;
```

```
u = double(t >= 0);
```

```
r = t .* double(t >= 0);
```

```
imp = double(t == 0); // ideal impulse approx is not possible in  
continuous
```

```
clf;
```

```
subplot(3,1,1);
```

```
plot(t, u);
```

```
xtitle("Unit Step (Continuous)");
```

```
subplot(3,1,2);
```

```
plot(t, r);
```

```
xtitle("Ramp (Continuous)");
```

```
subplot(3,1,3);
```

```
plot(t, imp);
```

```
xtitle("Impulse Approximation");
```

```
// Discrete time
```

```

n = -5:5;
ud = double(n >= 0);
rd = double(n .* (n >= 0));
impd = double(n == 0);
scf(1);
subplot(3,1,1);
plot2d3(n, ud);
xlabel("Unit Step (Discrete)");
subplot(3,1,2);
plot2d3(n, rd);
xlabel("Ramp (Discrete)");
subplot(3,1,3);
plot2d3(n, impd);
xlabel("Unit Impulse (Discrete)");

```

